Class09

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Background

In this mini-project, you will explore FiveThirtyEight's Halloween Candy dataset. FiveThirtyEight, sometimes rendered as just 538, is an American website that focuses mostly on opinion poll analysis, politics, economics, and sports blogging. They recently ran a rather large poll to determine which candy their readers like best.

Importing Candy Data

Let's get the data from the FiveThirtyEight GitHub repo. You can either read from the URL directly or download this candy-data.csv file and place it in your project directory.

```
candy_file <- "candy-data.csv"

candy <- read.csv(candy_file, row.names = 1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

Q1. How many different candy types are in this dataset?

```r
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```r
sum(candy$fruity)
```

```
[1] 38
```

What is Your Favorite Candy?

One of the most interesting variables in the dataset is winpercent. For a given candy this value is the percentage of people who prefer this candy over another randomly chosen candy from the dataset (what 538 term a matchup). Higher values indicate a more popular candy.

We can find the winpercent value for Twix by using its name to access the corresponding row of the dataset. This is because the dataset has each candy name as rownames (recall that we set this when we imported the original CSV file). For example the code for Twix is:

```r
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

An alternate way to do this is with the help of the dplyr package that we have seen previously:

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
     winpercent
Twix   81.64291
```

> Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```r
candy["Reese's Peanut Butter cup", ]$winpercent
```

```
[1] 84.18029
```

> Q4. What is the winpercent value for "Kit Kat"?

```r
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

> Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```r
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Side-note: the skimr::skim() function

There is a useful "skim()" function in the skimr package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

From your use of the "skim()" function use the output to answer the following:

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, the winpercent variable appears to be on a different scale than most of the other variables in the dataset. Most variables have values between 0 and 1, while winpercent has a mean around 50.3 and a much larger standard deviation.

> Q7. What do you think a zero and one represent for the candy$chocolate column?

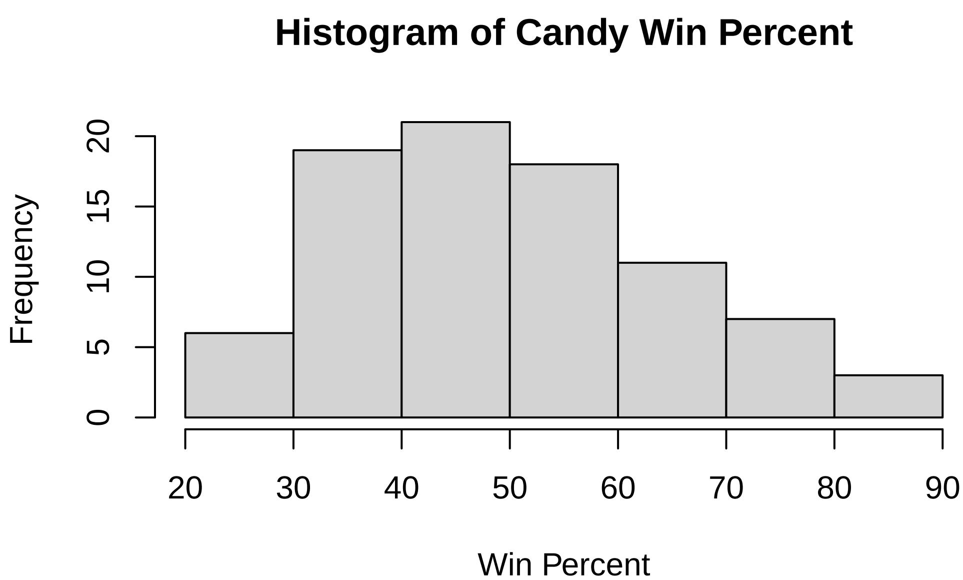
For the candy$chocolate column, a value of 1 means the candy contains chocolate, while a value of 0 means the candy does not contain chocolate.

Exploratory Analysis

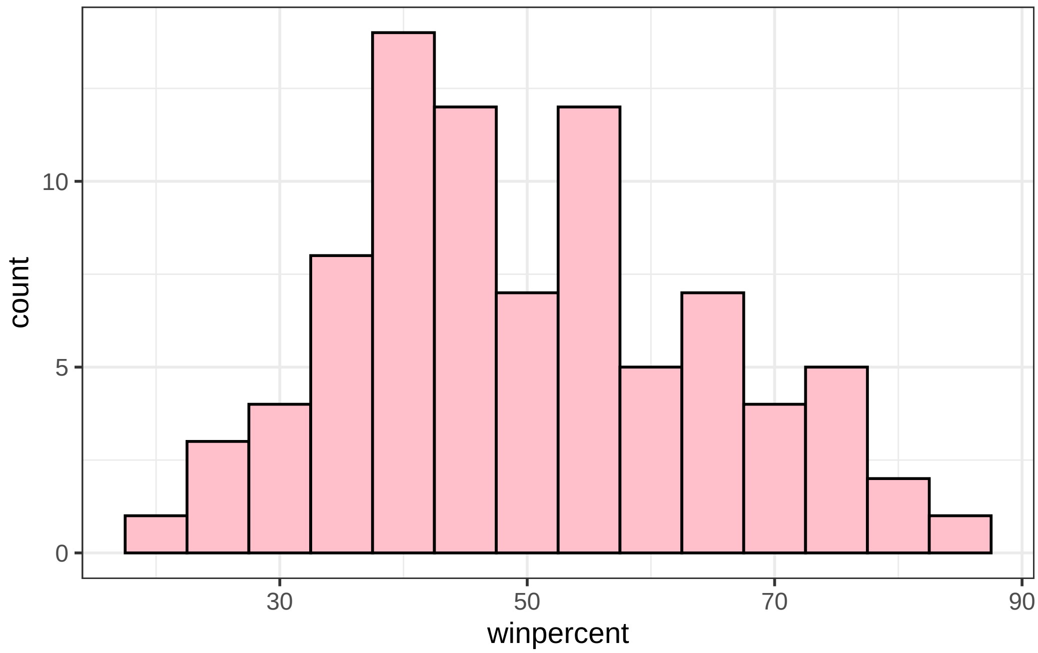
A good place to start any exploratory analysis is with a histogram. You can do this most easily with the base R function "hist()". Alternatively, you can use "ggplot()" with "geom_hist()".

> Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```r
hist(candy$winpercent,
     main = "Histogram of Candy Win Percent",
     xlab = "Win Percent")
```



```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(binwidth = 5, fill = "pink", color = "black") +
  theme_bw()
```



Q9. Is the distribution of winpercent values symmetrical?

The distribution of winpercent values is not perfectly symmetrical. It appears slightly right-skewed, with most candies clustered around the middle values and fewer candies having very high win percentages.

Q10. Is the center of the distribution above or below 50%?

The center of the distribution appears to be around or slightly above 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
chocolate <- candy$winpercent[as.logical(candy$chocolate)]

fruity <- candy$winpercent[as.logical(candy$fruity)]

mean(chocolate)
```

```
[1] 60.92153
```

```r
mean(fruity)
```

```
[1] 44.11974
```

On average, chocolate candy is ranked higher than fruity candy. Chocolate candies have an average winpercent of about 60.92, while fruity candies have an average winpercent of about 44.12.

> Q12. Is this difference statistically significant?

```r
t.test(chocolate, fruity)
```

```

	Welch Two Sample t-test

data:  chocolate and fruity
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y 
 60.92153  44.11974 
```

Yes, the difference is statistically significant. The t-test produced a very small p-value (2.871e-08), showing strong evidence that chocolate and fruity candies differ in their average winpercent values.

Overall Candy Rankings

Let's use the base R "order()" function together with "head()" to sort the whole dataset by a variable/column of interest. Or if you have been getting into the tidyverse and the dplyr package you can use the "arrange()" function together with "head()" to do the same thing and answer the following questions:

> Q13. What are the five least liked candy types in this set?

```r
head(candy[order(candy$winpercent), ], n = 5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

Q14. What are the top 5 favorite candy types?

```r
head(candy[order(candy$winpercent, decreasing = TRUE), ], n = 5)
```

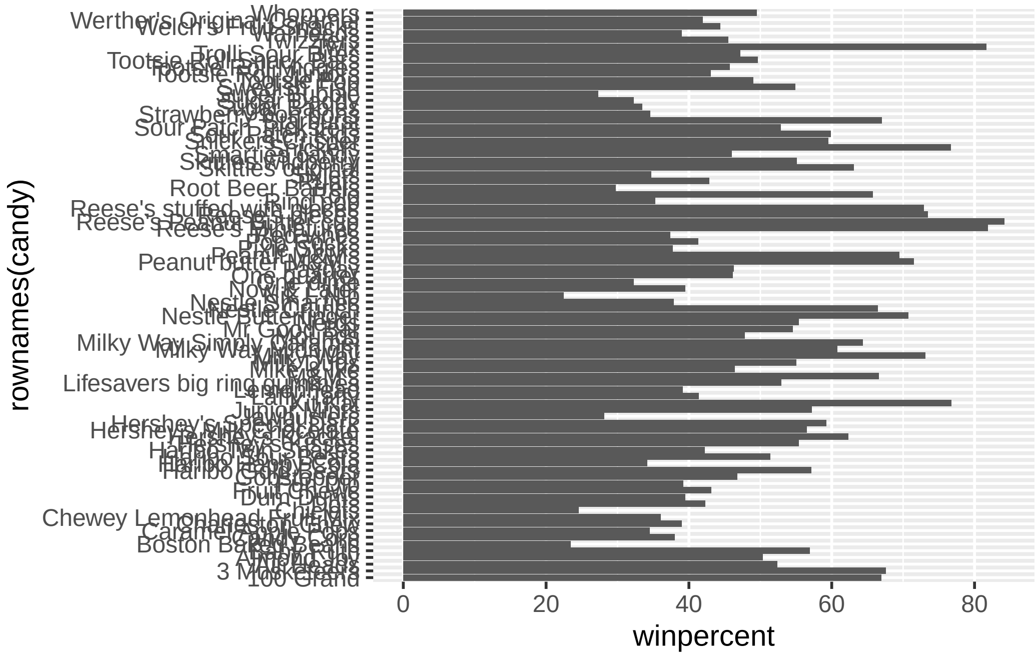
```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
```

```
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

To examine more of the dataset in this vain we can make a barplot to visualize the overall rankings.

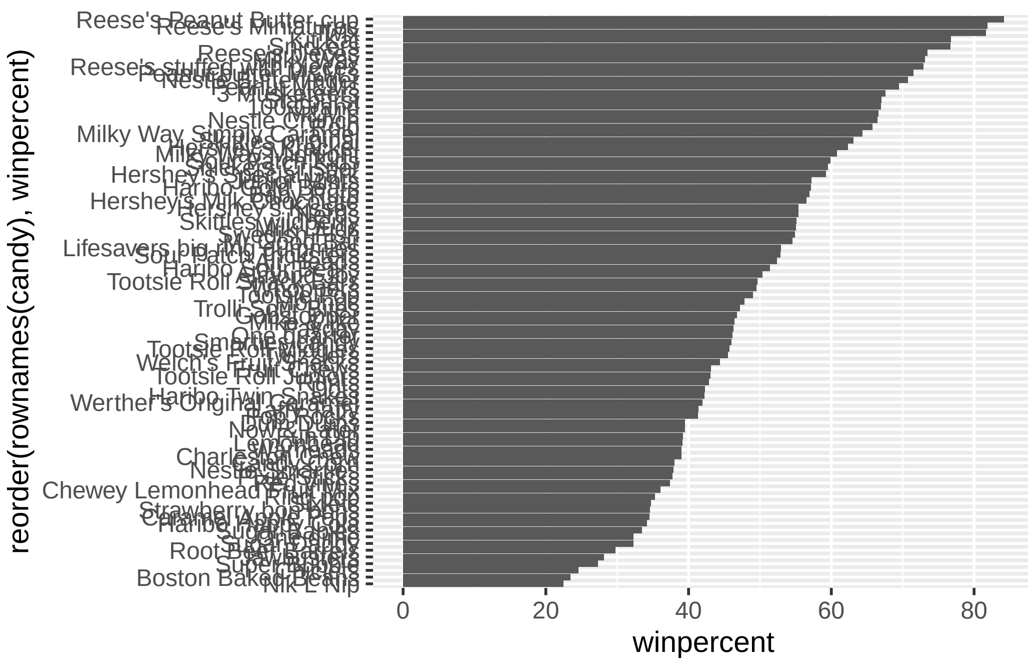
> Q15. Make a first barplot of candy ranking based on winpercent values.

```r
library(ggplot2)

ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



> Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```r
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```



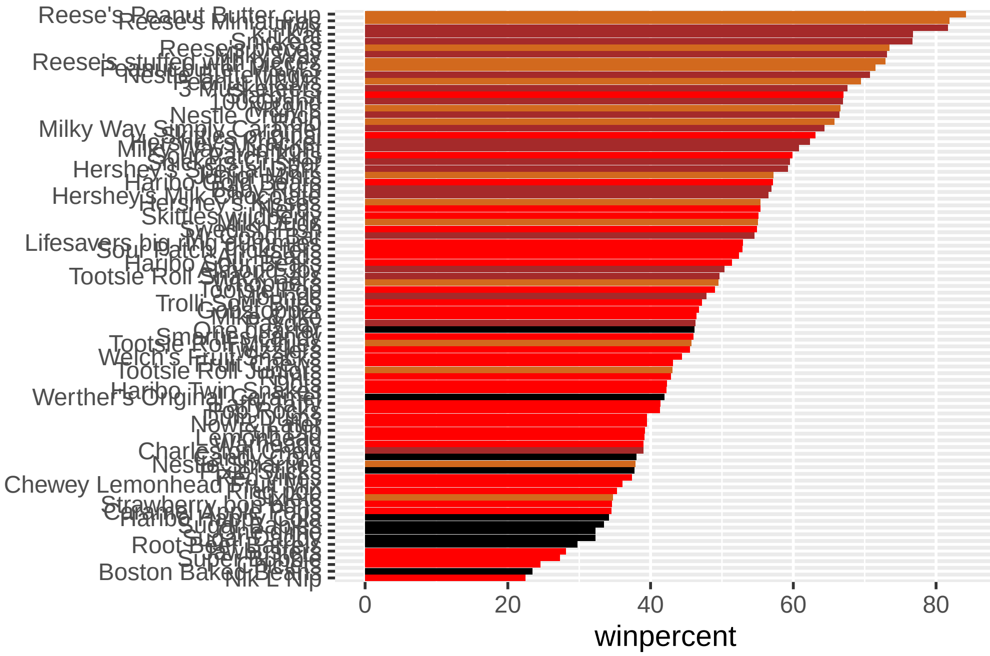
Time to add some useful color

Let's setup a color vector (that signifies candy type) that we can then use for some future plots.

```
my_cols = rep("black", nrow(candy))

my_cols[as.logical(candy$chocolate)] = "chocolate"

my_cols[as.logical(candy$bar)] = "brown"

my_cols[as.logical(candy$fruity)] = "red"
```

Now let's try our barplot with these colors.

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy appears to be Sixlets.

Q18. What is the best ranked fruity candy?

The best ranked fruity candy appears to be Starburst.

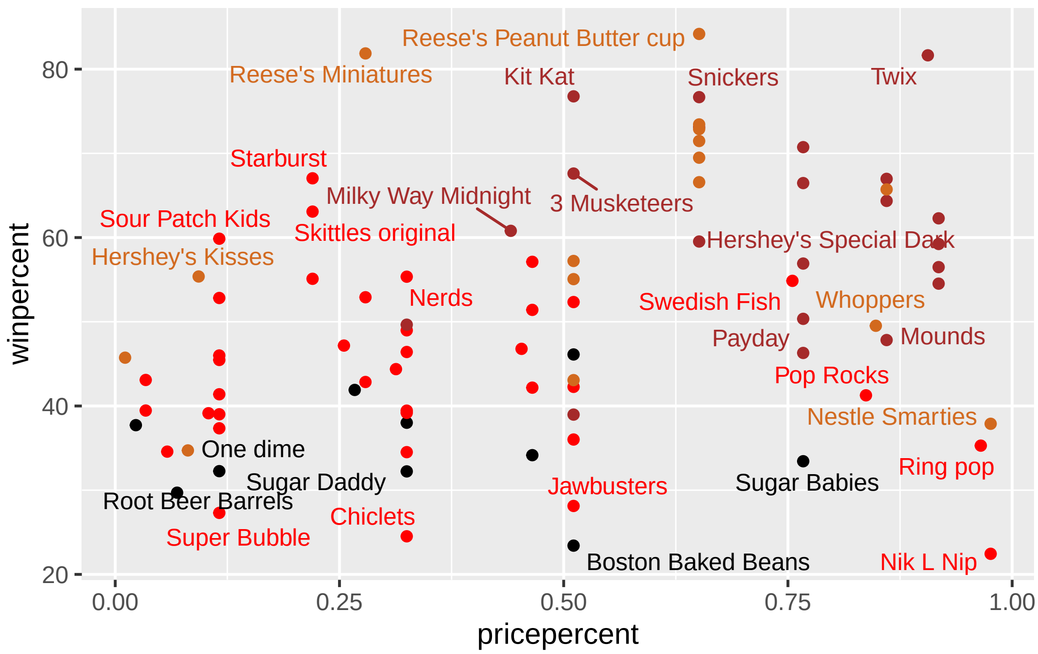
Taking a look at pricepercent

What about value for money? What is the best candy for the least money? One way to get at this would be to make a plot of winpercent vs the pricepercent variable. The pricepercent variable records the percentile rank of the candy's price against all the other candies in the dataset. Lower values are less expensive and higher values are more expensive.

To this plot we will add text labels so we can more easily identify a given candy. There is a regular "geom_label()" that comes with ggplot2. However, as there are quite a few candies in our dataset lots of these labels will be overlapping and hard to read. To help with this we can use the "geom_text_repel()" function from the ggrepel package.

```
library(ggrepel)

ggplot(candy) +
  aes(pricepercent, winpercent, label = rownames(candy)) +
```

```r
  geom_point(col = my_cols) +
  geom_text_repel(col = my_cols, size = 3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

The candy that appears to offer the most "bang for your buck" is Reese's Peanut Butter Cup, because it has one of the highest winpercent values while not being among the most expensive candies.

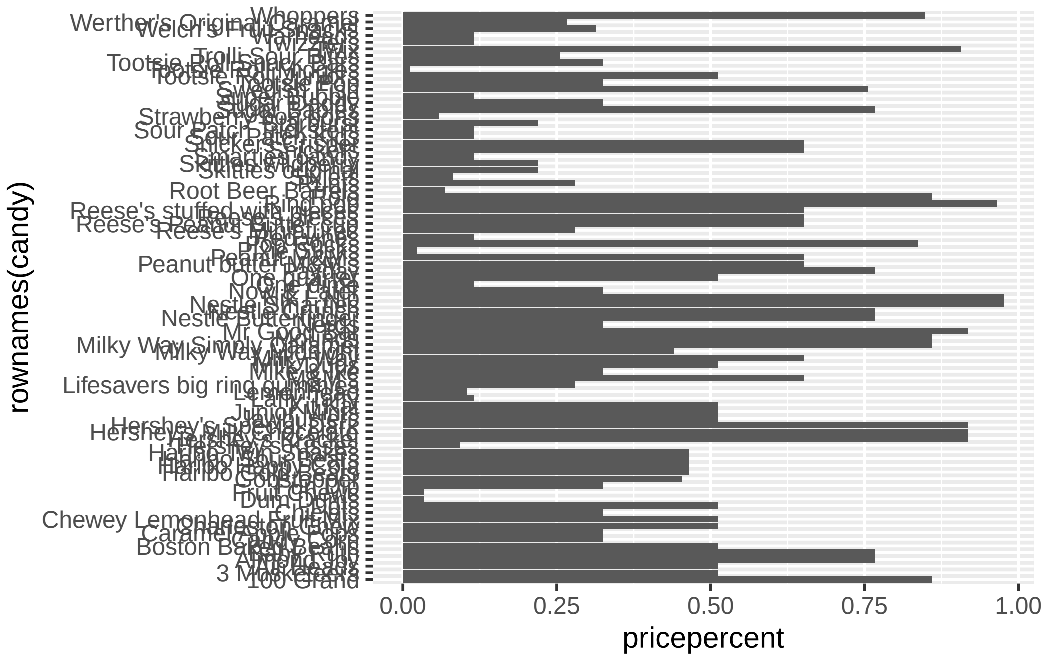
Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```r
ord <- order(candy$pricepercent, decreasing = TRUE)

head(candy[ord, c(11,12)], n = 5)
```

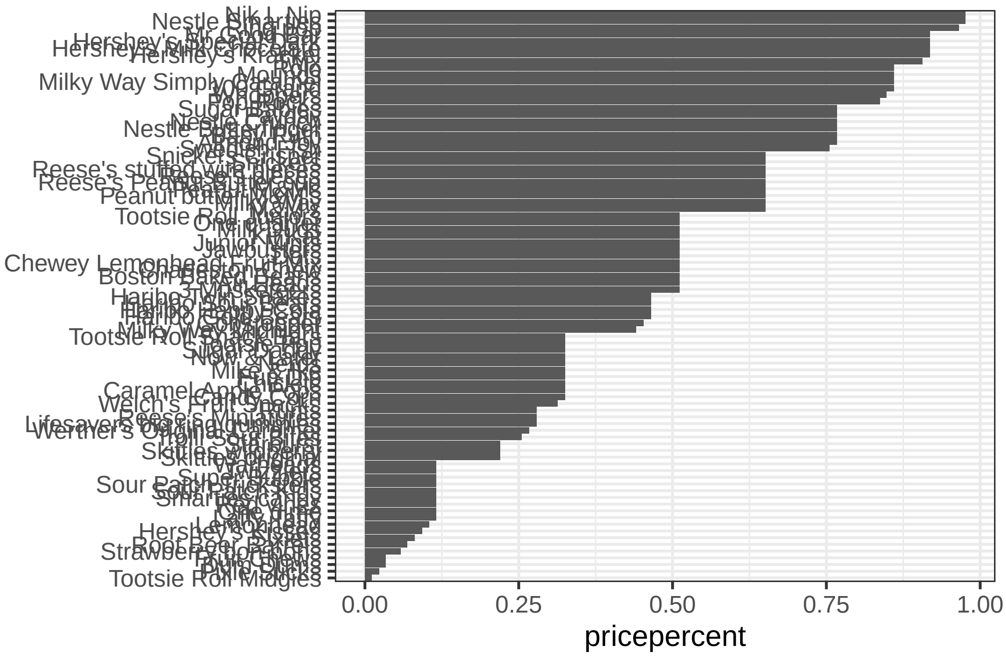
```
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
```

Q21. Make a barplot again with geom_col() this time using pricepercent and then improve this step by step, first ordering the x-axis by value and finally making a so called "dot chat" or "lollipop" chart by swapping geom_col() for geom_point() + geom_segment().

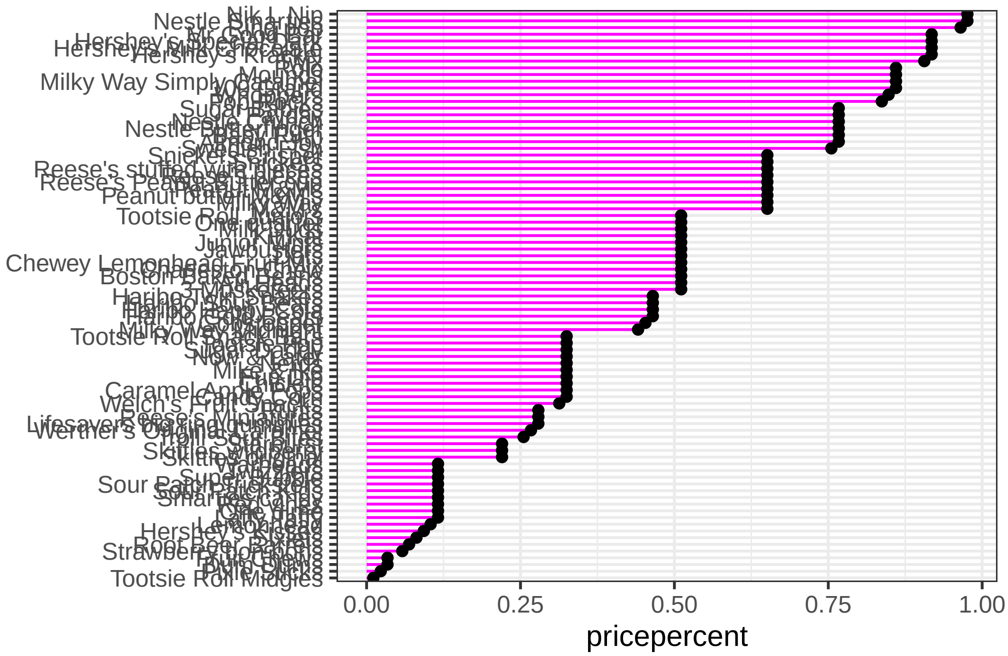
```
ggplot(candy) +
  aes(pricepercent, rownames(candy)) +
  geom_col()
```




```
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_col() +
  theme_bw() +
  ylab("")
```




```
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent),
                   xend = 0), col = "magenta") +
  geom_point() +
  theme_bw() +
  ylab("")
```



Exploring the correlation structure

Before we dive into PCA in the next section, it's worth pausing to look more closely at the correlations between all pairs of variables.

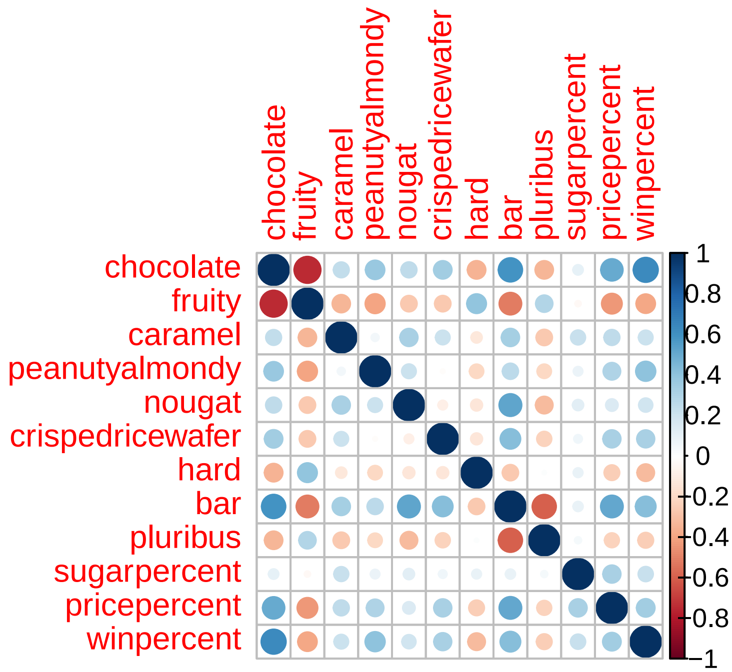
We have already seen that being a chocolate candy tends to imply being relatively expensive and popular - in other words, these variables are positively correlated with one another. Here we will examine the magnitude and direction of all variable pair combinations to see how they relate to each other more broadly.

The code below calculates the pairwise correlation between all variables and views the results with the corrplot package to plot a correlation matrix.

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



> Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The variables chocolate and fruity are strongly anti-correlated. This means candies that contain chocolate are generally not fruity, and fruity candies are generally not chocolate-based.

> Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Based on the corrplot, the variables most likely to have the largest contributions to PC1 are chocolate, fruity, bar, and peanutyalmondy. These variables show some of the strongest positive and negative correlations in the dataset and are likely to drive the major separation between different candy types in the PCA.

Principal Component Analysis

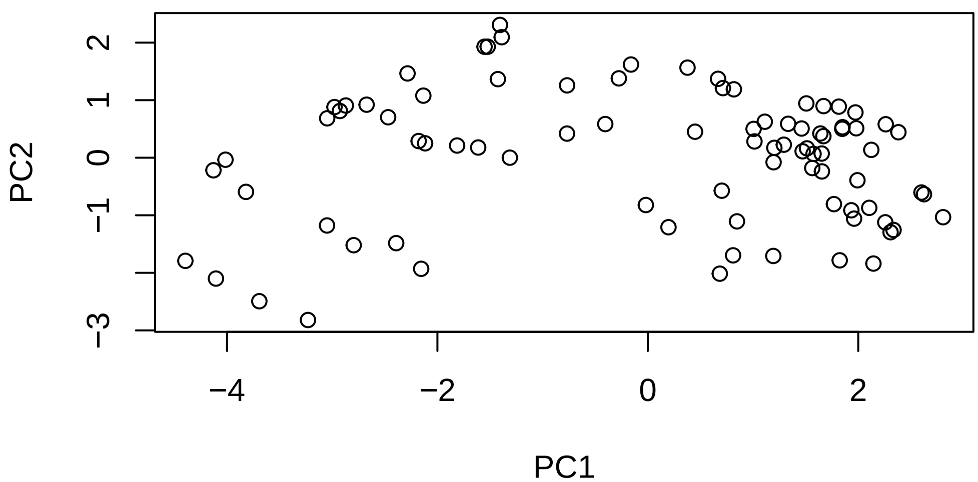
Let's apply PCA using the prcomp() function to our candy dataset remembering to set the scale=TRUE argument.

```
pca <- prcomp(candy, scale = TRUE)

summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

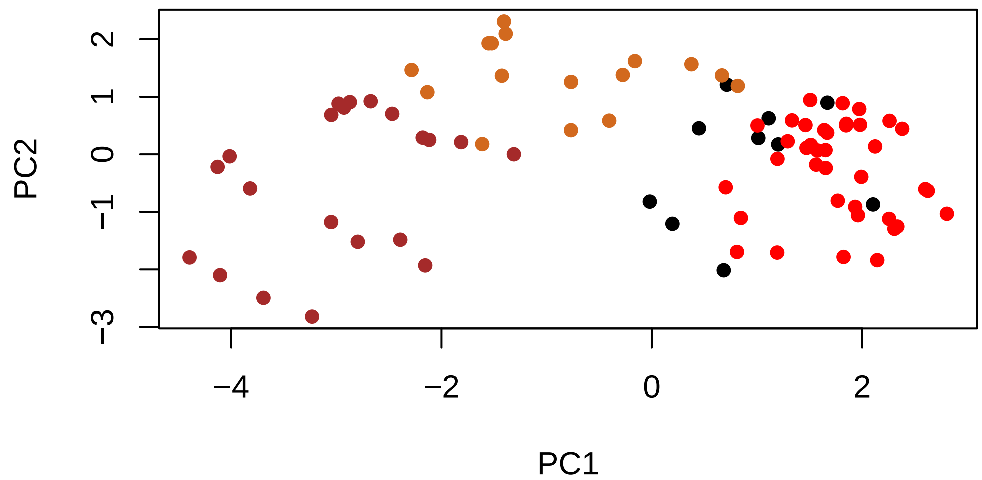
Now we can plot our main PCA score plot of PC1 vs PC2.

```
plot(pca$x[,1:2])
```



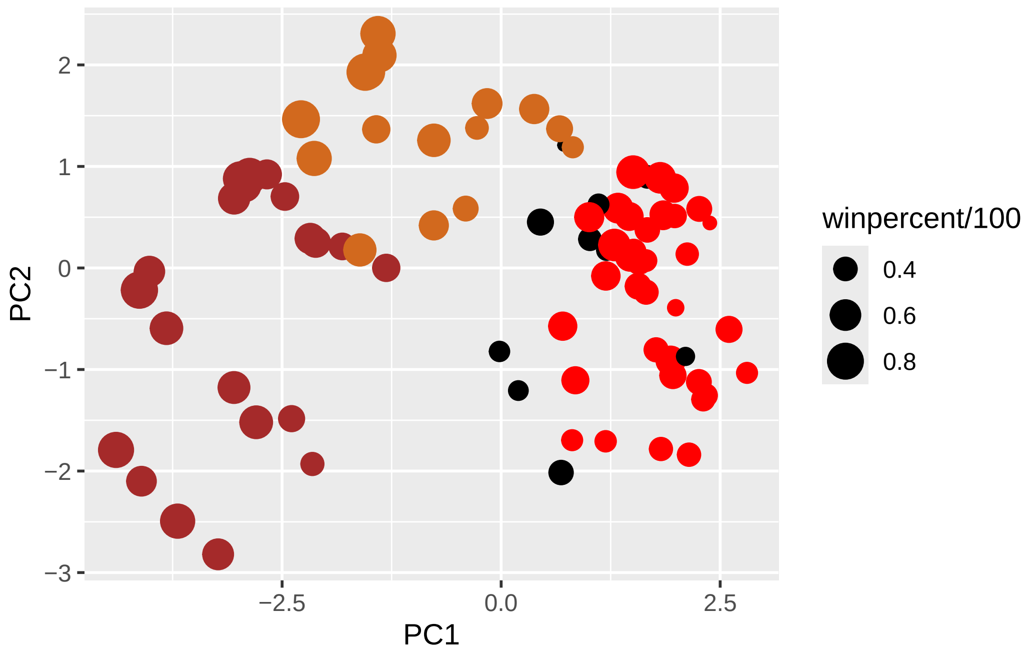
We can change the plotting character and add some color:

```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



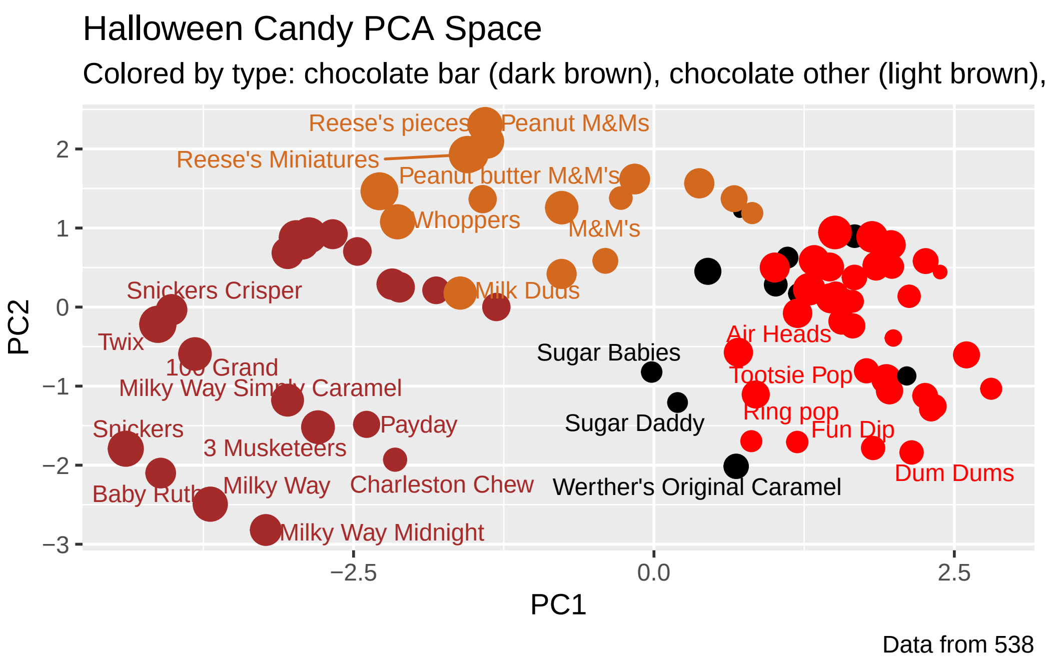
We can make a much nicer plot with the ggplot2 package but it is important to note that ggplot works best when you supply an input data.frame that includes a separate column for each of the aesthetics you would like displayed in your final plot. To accomplish this we make a new data.frame here that contains our PCA results with all the rest of our candy data. We will then use this for making plots below:

```
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
        aes(x = PC1, y = PC2,
            size = winpercent/100,
            text = rownames(my_data),
            label = rownames(my_data)) +
        geom_point(col = my_cols)

p
```



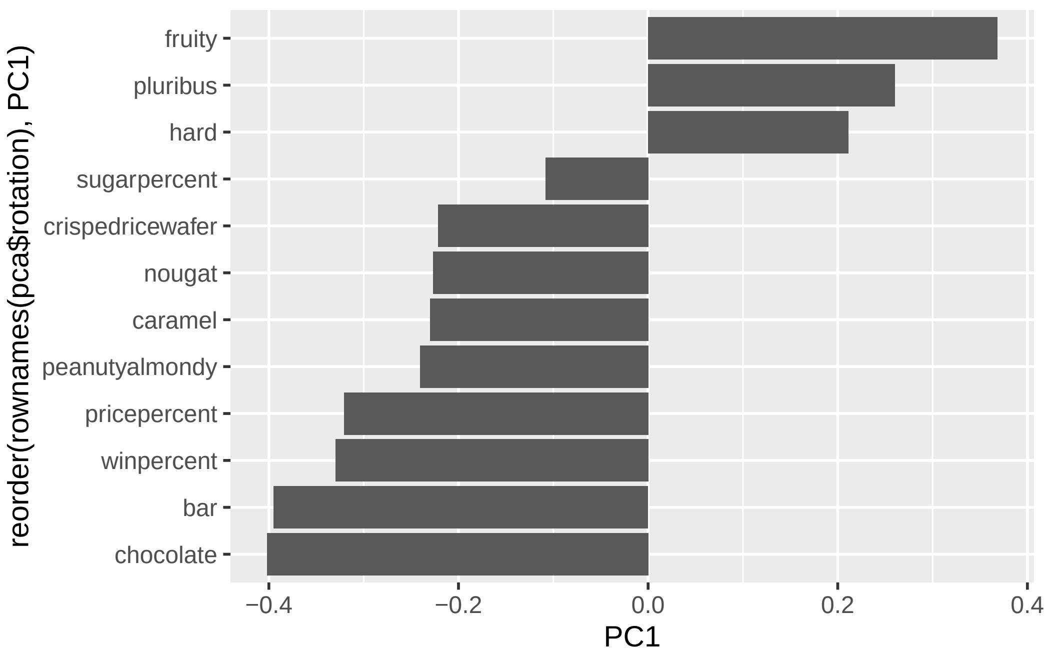
Again we can use the ggrepel package and the function ggrepel::geom_text_repel() to label up the plot with non overlapping candy names like. We will also add a title and subtitle like so:

```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7)  +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
       caption="Data from 538")
```



Let's finish by taking a quick look at PCA our loadings.

```
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



> Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables picked up strongly by PC1 in the positive direction are mainly fruity, hard, and pluribus. This makes sense because PC1 is separating fruity candies from chocolate/bar-type candies. We saw this relationship earlier in the corrplot, where chocolate and fruity were strongly anti-correlated.

Summary

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Candies that appear to be the most successful tend to be chocolate-based, often include peanut or caramel components, and are commonly candy bars. These candies generally had higher winpercent values in the exploratory plots and rankings. In contrast, fruity and hard candies tended to rank lower overall.

The different analyses supported each other well. The visualizations showed that many of the top-ranked candies were chocolate bars, while the correlation analysis revealed strong relationships between variables such as chocolate and bar, and an opposite relationship between chocolate and fruity. The PCA results reinforced these patterns by showing that chocolate/bar candies and fruity candies separated strongly along PC1. Together, these analyses consistently suggested that chocolate-related features are strongly associated with more popular "winning" candies.